

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287341

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

HIRAI; Yuta et al.

POSITION ESTIMATION SYSTEM

Abstract

According to one embodiment, a position estimation system includes a processor. The processor is configured to acquire first communication status data between a first wireless device and each of multiple second wireless devices, perform a first estimation process to estimate a position of the first wireless device using the first communication status data, and perform a second estimation process to estimate the position of the first wireless device using an output of a trained model. The first communication status data is acquired at different intervals depending on a channel. The first or second estimation process is selected based on the first communication status data. The trained model is built by using the training data including second communication status data.

Inventors: HIRAI; Yuta (Setagaya Tokyo, JP), ADACHI; Tomoko (Kawasaki Kanagawa, JP), AKITA; Koji (Yokohama Kanagawa, JP), NISHIKAWA; Masaki (Yokohama Kanagawa, JP), TANIGUCHI; Kentaro (Kawasaki Kanagawa, JP)

Applicant: KABUSHIKI KAISHA TOSHIBA (Tokyo, JP)

Family ID: 96950065

Assignee: KABUSHIKI KAISHA TOSHIBA (Tokyo, JP)

Appl. No.: 19/062149

Filed: February 25, 2025

Foreign Application Priority Data

JP	2024-034972	Mar. 07, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H04W64/00 (20090101); H04B17/318 (20150101); H04W76/15 (20180101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-034972, filed Mar. 7, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a position estimation system.

BACKGROUND

[0003] In recent years, for example, it has become common to estimate the position of a wireless device based on the strength of the signal (radio waves) observed between the device and another wireless device that has the function to perform wireless communication.

[0004] More specifically, the wireless device is a personal computer (hereafter referred to as PC) and, for example, estimating the position of the PC (employee) is considered to be useful in a case such as recognizing the position of people in an office building and efficiently controlling air conditioning.

[0005] However, the position of the above-described wireless device needs to be estimated at a high speed and with high accuracy.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a view showing an example of a configuration of a position estimation system according to an embodiment.

[0007] FIG. 2 is a block diagram showing an example of a functional configuration of a terminal device.

[0008] FIG. 3 is a block diagram showing an example of a functional configuration of a base station.

[0009] FIG. 4 is a block diagram showing an example of a functional configuration of a position estimation device.

[0010] FIG. 5 is a view showing an example of the system configuration of the position estimation device.

[0011] FIG. 6 is a flowchart showing an example of a procedure of the position estimation device.

[0012] FIG. 7 is a view illustrating an example of a method of obtaining a received signal strength.

[0013] FIG. 8 is a view illustrating another example of the method of obtaining the received signal strength.

[0014] FIG. 9 is a view illustrating yet another example of the method of obtaining the received signal strength.

[0015] FIG. 10 is a view illustrating yet another example of the method of obtaining the received signal strength.

[0016] FIG. 11 is a graph illustrating the accuracy of the position of the terminal device estimated by executing a first estimation process.

[0017] FIG. 12 is a view illustrating an overview of a trained model used in a second estimation process.

[0018] FIG. 13 is a view illustrating an overview of a trained model used in the second estimation

process.

DETAILED DESCRIPTION

[0019] In general, according to one embodiment, a position estimation system includes a processor. The processor is configured to acquire first communication status data related to a communication status between a first wireless device and each of multiple second wireless devices, perform a first estimation process to estimate a position of the first wireless device using the acquired first communication status data, perform a second estimation process to estimate the position of the first wireless device using an output of the trained model when the acquired first communication status data is input to the trained model. The first communication status data is acquired at different intervals depending on a channel connecting the first wireless device with each of the multiple second wireless devices. The first estimation process or the second estimation process is selected based on the first communication status data. The trained model is built by using the training data that includes second communication status data related to the communication status between the first wireless device and each of the multiple second wireless devices in a state in which the first wireless device is at a predetermined position.

[0020] Various embodiments will be described with reference to the accompanying drawings.

[0021] FIG. 1 shows an example of a configuration of a position estimation system according to an embodiment. The position estimation system 1 shown in FIG. 1 includes a terminal device 10, multiple base stations 20, and a position estimation device 30.

[0022] The terminal device 10 is an example of a wireless device including a function of executing wireless communication, and includes a notebook-type personal computer (notebook PC) used by, for example, a user (such as a company employee) who can move within a predetermined space such as an office building, but may also be the other electronic device such as a smartphone or may be a stationary device such as a desktop-type personal computer. In the embodiment, it is assumed that the terminal device 10 moves within a predetermined space together with the user.

[0023] The multiple base stations 20 are examples of wireless devices that include a function of executing wireless communication, and are arranged within the above-described predetermined space such as an office building. The terminal device 10 executes wireless communication with each of the multiple base stations 20 over a communication channel.

[0024] In this example, recognizing the position of the terminal device 10 used by a user moving in a predetermined space as described above (for example, a company employee moving inside an office building) corresponds to recognizing the position of the person (flow of person) in the space, and efficient air conditioning control, and the like in the space can be achieved by using data related to the position of the person (flow of person).

[0025] In this case, when it is considered that the communication status between the terminal device 10 and each of the multiple base stations 20 (hereinafter referred to as the communication status at the base station 20) depends on the distance between the terminal device 10 and the base station 20, it is thought that the position of the terminal device 10 can be estimated based on the communication status at each of the multiple base stations 20. Incidentally, in order to estimate the position of the terminal device 10 with high accuracy, it is desirable to use the communication status at a greater number of base stations 20.

[0026] However, the base stations 20 connected to the terminal device 10 over the communication channel are switched in accordance with the movement of the terminal device 10, and the terminal device 10 basically executes wireless communication with only one base station 20. For this reason, in order to recognize the communication status of the base stations 20 other than the base station 20 which executes wireless communication with the terminal device 10 over the communication channel, it is necessary to execute scanning (off-channel scanning) using an observation channel which is different from the communication channel.

[0027] In contrast, since actually necessary communication cannot be executed while scanning on the observation channel, executing scanning on the observation channel at a short interval (high

frequency) may be a factor which causes degradation in network performance. It is therefore necessary to execute scanning on the observation channel at a long interval (low frequency) (in other words, to reduce the time ratio of scanning on the observation channel).

[0028] However, if scanning on the observation channel is executed at a long interval (low frequency) as described above, it may take time to recognize the communication status at a certain number of base stations **20**, which is sufficient to estimate the position of the terminal device **10** with high accuracy, and it may not be possible to speed up the estimation of position of the terminal device **10**. In addition, if the position of the terminal device **10** is estimated without recognizing the communication status at a certain number of base stations **20**, which is sufficient to estimate the position of the terminal device **10** with high accuracy, the accuracy in estimation of the position may be degraded.

[0029] Therefore, the position estimation device **30** of the embodiment provides a mechanism for estimating the position of the terminal device **10** at high speed and with high accuracy, based on communication status information indicating the communication status between the terminal device **10** and each of the multiple base stations **20**.

[0030] Incidentally, in the embodiment, the communication status information will be described as, for example, received signal strength of packets (signals) observed between the terminal device **10** and each of the multiple base stations **20**, such as Received Signal Strength Indicator (RSSI).

[0031] In addition, in the embodiment, it is assumed that the terminal device **10** and each of the multiple base stations **20** execute wireless communication based on wireless LAN as specified in IEEE 802.11. In this case, the terminal device **10** corresponds to a station (STA), and each of the multiple base stations **20** corresponds to an access point (AP).

[0032] In IEEE 802.11k, a standard for the base stations **20** to scan observation channels other than the communication channels used for wireless communication in order to determine the status of use of radio waves is defined. Accordingly, for example, the communication between the terminal device **10** and a predetermined base station **20** can be observed by the other base stations **20** (i.e., the received signal strength at the other base stations **20** can be recognized).

[0033] Incidentally, in the embodiment, the status in which multiple base stations **20** are arranged (installed) at a relatively high density, in a predetermined space such as an office building is assumed, and it is assumed that even if (the user using) the terminal device **10** moves in the predetermined space, the received signal strength at each of the multiple base stations **20** can be measured using the communication channel and observation channel. Furthermore, as described above, the received signal strength at the base station **20** that executes wireless communication with the terminal device **10** over the communication channel (i.e., one of the multiple base stations **20**) can be measured (obtained) at a relatively short interval (high frequency) using the communication channel. In contrast, it is assumed that the received signal strength at the base station **20** other than the base station **20** that executes wireless communication with the terminal device **10** over the communication channel (i.e., the other of the multiple base stations **20**) is measured (acquired) at a relatively long interval (low frequency) using the observation channel as compared to the communication channel.

[0034] FIG. 2 is a block diagram showing an example of a functional configuration of the terminal device **10** shown in FIG. 1. As shown in FIG. 2, the terminal device **10** includes a transmission/reception unit **11** and a control unit **12**.

[0035] The transmission/reception unit **11** is a functional unit for transmitting and receiving various packets to and from each of the multiple base stations **20**. The control unit **12** is a functional unit for controlling the operation of the terminal device **10**.

[0036] FIG. 3 is a block diagram showing an example of a functional configuration of the base station **20** shown in FIG. 1. As shown in FIG. 3, the base station **20** includes a transmission/reception unit **21**, a control unit **22**, and an output unit **23**.

[0037] The transmission/reception unit **21** is a functional unit for transmitting and receiving various

packets to and from the terminal device **10**. The control unit **22** is a functional unit for controlling the operation of the base unit **20**. The output unit **23** is a functional unit for outputting the received signal strength (received signal strength at the base station **20**) of the packets observed using the communication channel or observation channel described above.

[0038] FIG. **4** is a block diagram showing an example of a functional configuration of the position estimation device **30** shown in FIG. **1**. As shown in FIG. **4**, the position estimation device **30** includes an acquisition unit **31**, a selection unit **32**, a first estimation unit **33**, a second estimation unit **34**, and a machine learning unit **35**.

[0039] The position estimation device **30** is connected to each of the multiple base stations **20** to execute wired or wireless communication. For example, the acquisition unit **31** acquires the received signal strength at the base station **20**, which is output from (the output unit **23** included in) each of the multiple base stations **20**. Incidentally, the interval (frequency) at which the received signal strength is output from each of the multiple base stations **20** (i.e., the interval at which the received signal strength is acquired by the acquisition unit **31**) differs depending on whether the received signal strength is the received signal strength of a packet observed using the communication channel or the received signal strength of a packet observed using the observation channel.

[0040] The selection unit **32** selects a first or second estimation process, based on the received signal strength acquired by the acquisition unit **31**.

[0041] When the first estimation process is selected by the selection unit **32**, the first estimation unit **33** executes the first estimation process. Incidentally, the first estimation process corresponds to the process of estimating the position of the terminal device **10** using the received signal strength acquired by the acquisition unit **31**.

[0042] When the second estimation process is selected by the selection unit **32**, the second estimation unit **34** executes the second estimation process. The second estimation unit **34** maintains a trained model to be described below. The second estimation process corresponds to a process of estimating the position of the terminal device **10** using the output of the trained model when the received signal strength acquired by the acquisition unit **31** is input to the trained model.

[0043] For example, the machine learning unit **35** learns the above-described trained model (i.e., builds the trained model) using the training data including the received signal strength at each of the multiple base stations **20** measured when the terminal device **10** is at the same position.

[0044] FIG. **5** is a view showing an example of a system configuration of the position estimation device **30** shown in FIG. **4**. The position estimation device **30** includes a CPU **301**, a nonvolatile memory **302**, a RAM **303**, a communication device **304**, and the like.

[0045] The CPU **301** is a processor to control the operation of various components in the position estimation device **30**. The CPU **301** may be a single processor or may be composed of multiple processors. The CPU **301** runs various programs loaded from the nonvolatile memory **302** into the RAM **303**. The programs which are run by the CPU **301** include an operating system (OS) and various application programs.

[0046] The nonvolatile memory **302** is a storage medium used as an auxiliary storage device. The RAM **303** is a storage medium used as a main storage device. FIG. **5** shows only the nonvolatile memory **302** and the RAM **303**. However, the position estimation device **30** may include, for example, the other storage devices such as a hard disk drive (HDD) and a solid state drive (SSD).

[0047] The communication device **304** is a device configured to execute wired communication or wireless communication.

[0048] Incidentally, in the embodiment, some or all of the units **31** to **35** shown in FIG. **4** can be realized by causing the CPU **301** (i.e., the computer of the position estimation device **30**) to run a predetermined program, i.e., by software. This program may be stored in a computer-readable storage medium and distributed or may be downloaded to the position estimation device **30** through a network. Incidentally, some or all of the units **31** to **35** may be realized by dedicated hardware, or

the like, or may be realized by combination of software and hardware.

[0049] An example of a procedure of the position estimation device **30** according to the embodiment will be described below with reference to a flowchart of FIG. **6**.

[0050] First, the acquisition unit **31** acquires the receiving signal strength at each of the multiple base stations **20** (step **S1**).

[0051] In the embodiment, the terminal device **10** can move in a predetermined space, and the terminal device **10** executes wireless communication over the communication channel with a base station **20** (hereinafter referred to as “first base station **20**”) among the multiple base stations **20**, depending on the position of the terminal device **10**. In this case, the received signal strength at the first base station **20** is measured at a short interval (high frequency) using the communication channel.

[0052] In addition, the received signal strength at each of the multiple base stations **20** (hereinafter referred to as “second base stations **20**”) other than the first base station **20** is measured at a long interval (low frequency) using the observation channel that is different from the communication channel.

[0053] In other words, in step **S1**, the received signal strength at the first base station **20** and the received signal strength at each of the multiple second base stations **20** are acquired at different intervals (timing). The received signal strength at each of the multiple base stations **20** (i.e., the first base station **20** and the multiple second base stations **20**), which is thus acquired in step **S1**, is stored (accumulated) in the position estimation device **30** in association with the time at which the received signal strength is acquired and an identifier for identifying the base station **20** each time the received signal strength is acquired.

[0054] Incidentally, in the embodiment, the multiple base stations **20** (first and second base stations **20**) are arranged (installed) at a relatively high density in the environment of an office building or the like, which is assumed to be a predetermined space. Therefore, even if the interval (frequency) of acquiring the received signal strength at each second base station **20** is low in such an environment, it can be expected that the interval between the acquisition of the received signal strength at any second base station **20** (packets are observed in any observation channel) is short.

[0055] In addition, the above-described received signal strength is acquired by unique methods provided by the manufacturers of the multiple base stations **20** (AP), methods based on the Simple Network Management Protocol (SNMP), or the like.

[0056] The method of acquiring the received signal strength of packets observed on the above-described observation channel will be described below. An example of wireless LAN under IEEE 802.11 will be described, and the standard for scanning the observation channel is assumed to be defined in IEEE 802.11k.

[0057] FIG. **7** is a view illustrating an example of a method of acquiring the received signal strength using the above-described observation channel.

[0058] In FIG. **7**, the second base station **20** observes packets between the terminal device **10** and the first base station **20** by scanning at a short time ratio over the observation channel (off-channel) **402** which is different from the communication channel (on-channel) **401** over which the first base station **20** executes wireless communication with the terminal device **10**, and measures the received signal strength of the packets (i.e., received signal strength at the second base station **20**).

[0059] The received signal strength at the second base station **20**, which is measured in this manner, is output from the second base station **20** to the position estimation device **30**, and is acquired by the position estimation device **30** (acquisition unit **31**).

[0060] Incidentally, for example, the second base station **20** can execute wireless communication with terminal devices other than the terminal device **10** over the communication channel **403**, during the time other than scanning on the observation channel **402**.

[0061] The method of acquiring the received signal strength using the observation channel has been described with reference to FIG. **7**. For example, however, the received signal strength may also be

acquired based on instructions from the first base station **20** (AP) to the terminal device **10** (STA). In this case, the terminal device **10** measures the received signal strength in response to the Beacon Measurement Request transmitted from the first base station **20**, and operates to reply with the measured received signal strength as a Beacon Measurement Report.

[0062] FIG. **8** shows another example of the method of acquiring the received signal strength using the observation channel. In this example, it is assumed that a mode (Measurement Mode) for measuring the received signal strength, which is set in advance, is active-all-ch or active-ch-rpt.

[0063] In FIG. **8**, first, the first base station **20** instructs (transmits) a Beacon Measurement Request to the terminal device **10** over the communication channel **401**.

[0064] Next, the terminal device **10** transmits a Probe Request to the second base station **20** over the observation channel (all or a specified channel) **402**, and thereby observes (receives) the Probe Response transmitted from the second base station **20**.

[0065] The terminal device **10** measures the received signal strength of the Probe Response (packet) thus observed as the received signal strength at the second base station **20**, and transmits the measured received signal strength to the first base station **20** as a Beacon Measurement Report.

[0066] The received signal strength thus transmitted from the terminal device **10** to the first base station **20** is output from the first base station **20** to the position estimation device **30** and is acquired by the position estimation device **30** (acquisition unit **31**).

[0067] FIG. **9** shows yet another example of the method of acquiring the received signal strength using the observation channel. In this example, it is assumed that Measurement Mode is Passive.

[0068] In FIG. **9**, first, the first base station **20** instructs (transmits) Beacon Measurement Request to the terminal device **10** over the communication channel **401**.

[0069] Next, for example, the terminal device **10** observes (receives) the beacon, which is periodically transmitted from the second base station **20**, by executing passive scanning over the observation channel (all or a specified channel) **402**.

[0070] The terminal device **10** measures the received signal strength of Beacon (packet) thus observed as the received signal strength at the second base station **20**, and transmits the observed received signal strength to the first base station **20** as Beacon Measurement Report.

[0071] The received signal strength thus transmitted from the terminal device **10** to the first base station **20** is output from the first base station **20** to the position estimation device **30** and is acquired by the position estimation device **30** (acquisition unit **31**).

[0072] Incidentally, in the example shown in FIG. **9**, unlike the above-described example shown in FIG. **8**, the terminal device **10** can observe Beacon transmitted from the second base station **20** without transmitting Probe Request.

[0073] The embodiment has been described with reference to FIG. **8** and FIG. **9**. For example, if the terminal device **10** manages a table holding (data of) the received signal strength of each of the multiple base stations **20** (first and second base stations **20**) that have already been measured using the communication channel and the observation channel, the terminal device **10** may operate to transmit the received signal strength held in the table as Beacon Measurement Report, to the first base station **20** (i.e., to return results of the received signal strengths stored in the terminal device **10** as a whole to the second base station **20**), in response to Beacon Measurement Request transmitted from the first base station **20**, as shown in FIG. **10**.

[0074] The received signal strength thus transmitted from the terminal device **10** to the first base station **20** is output from the first base station **20** to the position estimation device **30** and is acquired by the position estimation device **30** (acquisition unit **31**).

[0075] Incidentally, the operation described with reference to FIG. **10** is assumed to be executed when Measurement Mode is beacon-table.

[0076] As described above, the position estimation device **30** (acquisition unit **31**) of this embodiment can acquire the received strength at each of the multiple base stations **20**, but the acquisition of said received strength may also be achieved by other methods.

[0077] Next, for example, the position estimation device **30** executes a process of estimating the position of the terminal device **10** using the received signal strength at each of the above-described multiple base stations **20** at predetermined timing.

[0078] As a position estimation method using the received signal strength, for example, applying trilateration using the received signal strengths measured (observed) at three points is considered. However, it is possible to improve the accuracy of position estimation (i.e., to reduce the error term) by increasing the points where the received signal strength is measured.

[0079] In this example, the received signal strength at each of the multiple second base stations **20** other than the first base station **20** that executes wireless communication with the terminal device **10** over the communication channel can be acquired by executing scanning over the observation channel (i.e., observing packets over the observation channel) as described above.

[0080] In the embodiment, however, one terminal device **10** is described for convenience. It is assumed that the terminal devices other than the terminal device **10** exist in the above-described predetermined space of the office building, or the like, and that each of the multiple second base stations **20** executes wireless communication with the other terminal devices over the communication channel.

[0081] In this case, the second base station **20** cannot execute scanning over the observation channel at the same time as executing wireless communication over the communication channel. Therefore, if the time taken for the scanning over the observation channel increases, the time taken for the second base station **20** to execute wireless communication over the communication channel decreases and the network performance may be degraded.

[0082] For this reason, for example, it is desirable to make the ratio of the time for scanning over the observation channel to the time for executing wireless communication over the communication channel as small as possible. In such a case, the interval (frequency) with which the received signal strength is acquired by scanning over the observation channel is lowered.

[0083] In other words, when estimating the position of the terminal device **10** using the received signal strength (i.e., received signal strength at multiple points) at each of the multiple base stations **20**, it may take time to acquire the received signal strength at a sufficient number of base stations **20** (measurement points) in order to estimate the position of the terminal device **10** with high accuracy.

[0084] Therefore, in the embodiment, for example, the selection unit **32** maintains the conditions that is set in advance based on viewpoints such as whether the received signal strength is acquired to a degree to which the position of the terminal device **10** can be estimated with high accuracy. The estimation process selection conditions held in the selection unit **32** correspond to the conditions for selecting the first estimation process (first estimation unit **33**) or the second estimation process (second estimation unit **34**) (hereinafter referred to as “estimation process selection conditions”), and the selection unit **32** determines whether or not these estimation process selection conditions are met (step S2).

[0085] An example of the estimation process selection conditions used in step S2 will be described below.

[0086] First, it is considered that in order to estimate the position of the terminal device **10** with high accuracy as described above, the terminal device **10** needs to be located at a predetermined position during the period when the received signal strength at a sufficient number of base stations **20** is assumed to be measured (acquired).

[0087] For this reason, the above-described estimation process selection conditions include, for example, a condition that “the terminal device **10** is located at a predetermined position for a predetermined period (i.e., the time from the timing at which the terminal device **10** has moved to the present time is more than or equal to a certain time)” (hereinafter referred to as “first condition”). Incidentally, the situation that “the terminal device **10** is located at a predetermined position” is assumed to imply a case where the distance of movement is short enough to be

considered that the terminal device **10** has not moved.

[0088] If the terminal device **10** moves in a predetermined space (for example, an office building), the received signal strength at the first base station **20** that executes wireless communication with the terminal device **10** over the communication channel (i.e., the received signal strength of the packets observed over the communication channel) changes depending on the movement of the terminal device **10** (i.e., the distance between the terminal device **10** and the first base station **20**).

[0089] According to this, the timing at which the terminal device **10** has moved (i.e., whether or not the terminal device **10** has moved) can be determined based on the fact that the amount of change in the received signal strength of the packets observed over the communication channel among the received signal strengths stored in the position estimation device **30** is greater than or equal to a predetermined value (i.e., the received signal strength has increased or decreased by a certain amount or more).

[0090] Incidentally, the timing at which the terminal device **10** has moved may be determined based on the fact that the amount of change in the received signal strength observed over the predetermined observation channel is greater than or equal to a certain value.

[0091] In addition, if the terminal device **10** (for example, a notebook PC) is started up in a predetermined space, the acquisition of the received signal strength is executed immediately after the terminal device **10** is started up (i.e., the received signal strength at the position of the terminal device **10** is newly measured). Thus, the situation in which the received signal strength is newly measured can be considered to be the same as that when the terminal device **10** has moved.

[0092] For this reason, if the interval in the acquisition of the received signal strengths of the packets observed over the same channel (communication channel or observation channel) among the received signal strengths stored in the position estimation device **30** is a certain time or more, it may be determined as the timing when the terminal device **10** has moved (started up).

[0093] The case where the terminal device **10** is started up has been described. However, the conditions are the same as those in a case where the terminal device **10** that has already been started up enters a predetermined space, or the like.

[0094] Furthermore, when the terminal device **10** moves in a predetermined space, the base station **20** to which the terminal device **10** is connected over the communication channel (i.e., executes the wireless communication) may be changed (switched).

[0095] According to this, the timing at which the terminal device **10** has moved may be determined based on the fact that the base station **20** (AP) to which the terminal device **10** is connected over the communication channel has been changed.

[0096] Furthermore, in order to estimate the position of the terminal device **10** with high accuracy, it is desirable that the received signal strength at a sufficient number of base stations **20** is acquired.

[0097] For this reason, the above-described estimation process selection conditions may include a condition such as “the number of base stations **20** for which the received signal strength has been acquired is a certain number or more” (hereinafter referred to as “second condition”). Incidentally, the number of base stations **20** for which the received signal strength has been acquired can be acquired by counting the number of base stations **20** for which the received signal strength has been acquired from the timing at which the above-described terminal device **10** has moved to the present time with reference to the received signal strength stored in the position estimation device **30**.

[0098] Incidentally, the estimation process selection conditions may be at least one of the first and second conditions, or may be conditions obtained by partially modifying some parts of the first or second conditions.

[0099] Furthermore, the estimation process selection conditions may be conditions prepared by combining the first and second conditions. More specifically, according to the estimation process selection conditions obtained by combining the above-described first and second conditions, whether or not the received signal strengths at a sufficient number of base stations **20** have been acquired when the terminal device **10** is located at a predetermined position can be determined.

[0100] Incidentally, the estimation process selection conditions in the embodiment may be any conditions set based on the viewpoint such as whether the received signal strength is acquired such that the position of the terminal device **10** can be estimated with high accuracy as described above, and are not limited to those described here.

[0101] If it is determined that the estimation process selection conditions are met (step S2: YES), the selection unit **32** selects the first estimation process (first estimation unit **33**) (step S3).

[0102] When the process in step S3 is executed, the first estimation unit **33** executes the first estimation process (step S4).

[0103] In contrast, if it is determined that the estimation process selection conditions are not met (NO in Step S2), the selection unit **32** selects the second estimation process (second estimation unit **34**) (step S5).

[0104] When the process in step S5 is executed, the second estimation unit **34** executes the second estimation process in step S4.

[0105] According to the above-described processes shown in FIG. 6, the first estimation process can be executed if, for example, the terminal device **10** remains at the same position for a certain period or longer and the received signal strengths at a sufficient number of base stations **20** are acquired (observed), and the second estimation process can be executed if, for example, the received signal strengths at a sufficient number of base stations **20** are not acquired (observed) immediately after the terminal device **10** moves (i.e., the first and second estimation processes can be switched based on the estimation process selection conditions).

[0106] Incidentally, in the embodiment, the first or second estimation process can be selected (i.e., the first and second estimation processes can be switched) based on the received signal strength at each of the multiple base stations **20** stored in the position estimation device **30**. For example, however, if the received signal strength described with reference to FIG. 10 is acquired in step S1, the acquired received signal strength may include, for example, the received signal strength measured before the timing at which the terminal device **10** has moved previously. In this case, in the embodiment, when the terminal device **10** moves as described above, the second estimation process is selected (i.e., the first estimation process is switched to the second estimation process). Therefore, by retaining (the information indicating) the timing at which the estimation process is switched in the position estimation device **30**, only the received signal strength acquired after that timing (i.e., the received signal strength acquired after the terminal device **10** has moved) may be used when the first or second estimation process is selected. The received signal strengths used when selecting the first or second estimation process have been described. However, the received signal strengths used in the first and second estimation processes to be described below are the same as those described above.

[0107] The first and second estimation processes in the embodiment will be described below. First, the first estimation process is executed when the terminal device **10** remains at the same position for a sufficiently long time, as described above, and the received signal strength at each of all base stations **20** located in the vicinity of the position (i.e., base stations **20** located near the terminal device **10**) is acquired. In the first estimation process, the received signal strength acquired from the timing when the terminal device **10** has moved until the present time is used among the received signal strengths stored in the position estimation device **30** (i.e., the received signal strengths acquired by the acquisition unit **31**).

[0108] In general, a relationship between the received signal strength (RSSI) and the distance is expressed by the following formula (1). Incidentally, to simplify the formula, the only received signal strength in the 2.4 GHz band is focused.

$$[00001] \text{ rssi}_i = A - 10 * n_i * \log_{10}(\sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z^2)}) \quad \text{Formula(1)}$$

[0109] In formula (1), i represents an index assigned to each base station **20** (AP), rssi.sub.i represents the received signal strength of the base station **20** whose index is i (hereafter referred to

as base station **20-i**), and $x_{\text{sub},i}$, $y_{\text{sub},i}$, and $z_{\text{sub},i}$ represent the position (x , y , and z coordinate values) of the base station **20-i**. In addition, in formula (1), x , y , and z represent the position of the terminal device **10**. Furthermore, in formula (1), A represents the received signal strength when the distance between the terminal device **10** and the base station **20-i** is 1 m, which is different from for each terminal device **10**. Incidentally, since A represents the received signal strength when the distance between the terminal device **10** and the base station **20-i** is sufficiently short, it can be said that the received signal strength is equivalent to the transmitted signal strength of the terminal device **10**. In addition, in formula (1), $n_{\text{sub},i}$ represents an attenuation constant, and $n_{\text{sub},i}=2$ in an ideal free space. However, it is known that $n_{\text{sub},i}$ fluctuates in a complex manner depending on the surrounding conditions in an indoor environment, the positions of the terminal device **10** and the base station **20-i**, and the like, and $n_{\text{sub},i}$ takes a value between 2 and 6 in many cases.

[0110] Incidentally, in the embodiment, the position of each of the multiple base stations **20** is assumed to be held (registered) in advance, and the values of $x_{\text{sub},i}$, $y_{\text{sub},i}$, and $z_{\text{sub},i}$, which indicate the position of the above-described base station **20-i** are known. In addition, if the terminal device **10** is a notebook PC and is placed on a desk or the like for use, the position of the terminal device **10** in the height direction can be expected to be approximately 1 m, and z is also assumed to be known. Furthermore, if all $n_{\text{sub},i}$ are assumed to take the same value n , unknown variables in formula (1) are four variables, i.e., A , n , x , and y .

[0111] In this case, the above-mentioned unknown variables A , n , x , and y can be obtained by solving the minimization problem expressed by the following formula (2).

$$[00002] \text{ minimize } \text{Math.}_{i \in \text{AP}} (d_{\text{rssi},i} - \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2})^2 \quad \text{Formula(2)}$$

where $d_{\text{rssi},i} = 10^{\frac{A - \text{rssi}_i}{10n}}$

[0112] In formula (2), $d_{\text{sub.rssi},i}$ represents the distance between the terminal device **10** and the base station **20-i**, which is calculated from $\text{rssi}_{\text{sub},i}$ (i.e., the received signal strength at the base station **20-i**) described above.

[0113] In the embodiment, the x , y and the above-mentioned known z (i.e., x , y , and z) obtained by solving such a minimization problem correspond to the position of the terminal device **10**, which is the estimation result of the first estimation process.

[0114] Incidentally, when the minimization problem expressed by the above-mentioned formula (2) is solved, a constraint condition that the position of the terminal device **10** is in a predetermined space (i.e., an office building) may be added as needed. Furthermore, A may be used to represent a fixed value by using the knowledge that the transmission power of the terminal device **10** (for example, a notebook PC) is generally the same.

[0115] FIG. **11** is a box plot diagram with the vertical axis showing an error between the position (estimated value) of the terminal device **10** estimated by executing the first estimation process and an actual position (ground truth) of the terminal device **10**, and a horizontal axis showing the number of (received signal strengths at) base stations **20** used in the first estimation process.

[0116] It is shown in FIG. **6** that the error between the estimated value and the ground truth becomes smaller as the number of base stations **20** increases, and it can be understood that the accuracy in estimation of the position of the terminal device **10** is improved.

[0117] As described above, in the first estimation process, it is considered that if the terminal device **10** is sufficiently stationary and the received signal strengths (RSSI) of a large number of base stations **20** are acquired, relatively high-accuracy position estimation can be executed.

[0118] Incidentally, the first estimation process described here is just one example, and the other processes may be executed as long as they estimate the position of the terminal device **10** using the received signal strengths at a sufficient number of base stations **20**, which are acquired during the period when the terminal device **10** is located at a predetermined position.

[0119] Incidentally, as described above, A , which is used to estimate the position of the terminal device **10** in the first estimation process, is the received signal strength at a distance of 1 m between

the terminal device **10** and the base station **20-i**, and is equivalent to the transmitted signal strength of the signal transmitted from the terminal device **10**. In the embodiment, A may be an unknown constant. However, since A is generally determined according to, for example, the type of the terminal device **10** such as a notebook PC or smartphone, A may be determined according to the type of terminal device **10** and applied to formula (2). According to this, A is set based on the type of terminal device **10**, and the optimization problem can be solved with fewer unknowns (the solution space is reduced, which allows for higher accuracy in estimation).

[0120] Incidentally, the type of terminal device **10** can be estimated by, for example, using the vendor ID included in the MAC address used in wireless communication, or by using information on whether or not mobile communication can be executed, which is included in Probe Request from the terminal device **10** (i.e., information on the type of terminal device **10**), and the like.

[0121] In addition, it has been described that the first estimation process is executed by targeting only the received signal strength in the 2.4 GHz band. However, the propagation characteristics differ depending on the frequency band of the radio waves used for wireless communication. In wireless LAN, two frequency band, 2.4 GHz band and 5 GHz band, are mainly used. In this case, the first estimation process may be executed by distinguishing between the 2.4 GHz band and the 5 GHz band. The first estimation process has been described here. However, the second estimation process may be executed in the same manner.

[0122] Furthermore, when each of the multiple base stations **20** includes multiple antennas, the multiple antennas arranged at the same base station **20** have different characteristics depending on the direction and position of the antennas. For this reason, for example, if one base station **20** includes first and second antennas and packets are observed using each of the first and second antennas, the received signal strength of the packets observed using the first antenna and the received signal strength of the packets observed using the second antenna may be distinguished (i.e., treated as separate received signal strengths) in the first estimation process. The first estimation process has been described here. However, the second estimation process may be executed in the same manner.

[0123] In addition, as described above, when using Beacon Measurement Report, the terminal device **10** measures the received signal strength. However, if the transmitted signal strength of each base station **20** (i.e., the transmitted signal strength of the signal transmitted from the base station **20**) is known, a value obtained by subtracting the received signal strength from the transmitted signal strength (i.e., the difference between the transmitted signal strength and the received signal strength) becomes an attenuation term of the radio waves. In other words, when using the value obtained by subtracting the received signal strength from the transmitted signal strength, the first estimation process can be executed using a formula which does not require A.

[0124] Incidentally, in the above formula (2), it has been described that the position of (multiple base stations **20** including) the base station **20-i** is known. However, (the information indicating) the position of each of the base stations **20** may be acquired from the base station **20** via a wireless LAN controller (wireless device), for example, based on SNMP or the like. In addition, if the base station **20** responds to Location Configuration Information (LCI) request (i.e., if a base station **20** that supports LCI is used), the position estimation device **30** may acquire the position of the base station **20** by means of the LCI request.

[0125] In addition, if the position of each of the multiple base stations **20** cannot be acquired as described above, for example, drawings for installation constructions of the base station **20** or the position input (indicated) by a user who has visually confirmed the position of the base station **20** arranged in a predetermined space may be used.

[0126] In some cases, however, the base station **20** cannot be arranged (installed) as indicated by the above-mentioned drawings for installation construction or even if the position based on the drawings or the like is input, the position may be different from the actual position of the base station **20**. In this case, for example, the received signal strength at each of the multiple base

stations **20** may be measured by transmitting and receiving the packets between the multiple base stations **20**, the position of the predetermined base station **20** may be estimated using the received signal strength, and the position input as described above may be corrected based on the estimated position (i.e., the input position may be corrected to the estimated position). Incidentally, when estimating the position of the base station **20** using the received signal strength, for example, the same process as the above-described first estimation process may be executed.

[0127] Next, the second estimation process will be described. As described above, the first estimation process is executed when the received signal strengths at a sufficient number of base stations **20** are acquired in the state in which the terminal device **10** is located in a predetermined position (i.e., stays at the same position) while the second estimation process is executed when the received signal strengths at a sufficient number of base stations **20** are not acquired. Incidentally, in the second estimation process, similarly to the first estimation process, the received signal strength acquired from the timing when the terminal device **10** has moved until the present time is used among the received signal strengths stored in the position estimation device **30** (i.e., the received signal strengths acquired by the acquisition unit **31**).

[0128] The number of received signal strengths used in the second estimation process is not sufficient to estimate the position of the terminal device **10** with high accuracy. However, if the same process as the first estimation process is executed using only these received signal strengths, the accuracy in results of the estimation of the position of the terminal device **10** is considered to be low.

[0129] Therefore, in the second estimation process, for example, the position of the terminal device **10** is estimated using a trained model built by applying a known machine learning algorithm such as a neural network.

[0130] An outline of the trained model used in the second estimation process will be described below with reference to FIG. **12** and FIG. **13**.

[0131] First, it is assumed that, for example, the received signal strengths at all base stations **20** in the vicinity of the terminal device **10** are acquired and accumulated by keeping the terminal device **10** stationary (i.e., at a predetermined position) for a sufficient period of time at the same position in a predetermined space, before the operation of the position estimation system **1**.

[0132] In this case, in the embodiment, by using the training data that includes the received signal strengths and the ground truth at all base stations **20**, which have been accumulated as described above (i.e., by learning from the training data), a trained model that outputs the ground truths when the received signal strengths are input is built (constructed).

[0133] Incidentally, in the embodiment, the ground truths are assumed to be the received signal strengths at all base stations **20** in the vicinity of the terminal device **10**, which have been accumulated as described above.

[0134] FIG. **12** shows a concept of the trained model which is trained using the received signal strengths at all base stations **20** as described above.

[0135] In FIG. **12**, it is assumed that N base stations **20** are arranged in a predetermined space, and the received signal strength at base station $20-i$ ($i=1, 2, \dots, j, \dots, N$) among N base stations **20** is represented by $AP.sub.iRSSI$, and whether or not the received signal strength has been acquired is represented by $AP.sub.iused$. Incidentally, when $AP.sub.iused=1$, it is indicated that the received signal strength at base station $20-i$ is acquired, and when $AP.sub.iused=0$, it is indicated that the received signal strength at base station $20-i$ is not acquired.

[0136] Incidentally, for example, if the received signal strength at base station $20-i$ is acquired multiple times while the terminal device **10** is moved from the predetermined position to the same position, the received signal strength at base station $20-i$ included in the training data may be the received signal strength which is subjected to preprocessing such as calculating the average of the received signal strengths acquired multiple times.

[0137] In other words, FIG. **12** shows the learning phase of constructing a trained model (neural

network) capable of outputting the received signal strengths (estimated values) at all base stations **20-1** to **20-N** when the received signal strengths at all base stations **20-1** to **20-N** are input.

[0138] Furthermore, if it is assumed that the received signal strengths at all base stations **20** are accumulated as described above, in the embodiment, a trained model of outputting the received signal strengths (i.e., ground truths) at all base stations **20** when the received signal strengths are input is built by using the training data including the received signal strengths at some base stations **20** among all the base stations **20** (i.e., by learning from the received signal strengths at some base stations **20**).

[0139] Incidentally, the received signal strengths at some base stations **20**, which are used as the training data, may be extracted intentionally or randomly from the received signal strengths at all the base stations **20**.

[0140] FIG. **13** shows a concept of the trained model which is trained using the received signal strengths at some base stations **20** as described above.

[0141] In FIG. **13**, it is shown that AP.sub.jused=-1 and that the received signal strength at base station **20-j** is not used as the training data.

[0142] In other words, FIG. **13** shows the learning phase of constructing a trained model (neural network) capable of outputting the received signal strengths (estimated values) at all base stations **20-1** to **20-N** when the received signal strengths at some base stations **20** except, for example, at least the base station **20-j** among all the base stations **20-1** to **20-N** are input.

[0143] By performing the learning described above with reference to FIG. **12** and FIG. **13**, the trained model capable of outputting the received signal strengths at all base stations **20** is built even when only the received signal strengths at a limited number of base stations **20** are input.

[0144] In this case, it has been described that the learning is executed using the received signal strength acquired when the terminal device **10** is located at a predetermined position (stationary at the same position). However, a high-accuracy trained model can be generated by repeating the learning for each position of the terminal device **10**.

[0145] In addition, the processing of generating the trained model in the embodiment (i.e., the processing related to training of the trained model) is assumed to be executed by the machine learning unit **35** included in the position estimation device **30**. However, it may also be executed outside the position estimation device **30**.

[0146] As described above, when the second estimation process is executed in the embodiment, the received signal strengths at a sufficient number of base stations **20** are not obtained (i.e., only the received signal strengths at some of the base stations **20** are obtained). However, the second estimation unit **34** can acquire the received signal strengths at all of the base stations **20**, which are output from the trained model, by inputting the received signal strengths at some of the base stations **20** to the above-described trained model.

[0147] Since the received signal strengths at all the base stations **20** acquired in this manner correspond to the received signal strengths at a sufficient number of base stations **20**, which are acquired when the above-described terminal device **10** is located at a predetermined position, the second estimation unit **34** can estimate the position of the terminal device **10** with high accuracy by executing the same process as the above-described first estimation process using the acquired received signal strengths (i.e., the output of the trained model).

[0148] In the embodiment, the trained model that is generated to output the received power at all the base stations **20** by inputting the received power at some base stations **20** has been described. For example, however, the trained model may also be generated to output the position (estimated value) of the terminal device **10** by inputting the received power at some base stations **20**.

[0149] Such a trained model is built by learning from the training data which includes the received signal strengths at all the base stations **20**, which are acquired and accumulated as described above, the received signal strengths at some of the base stations **20** among all the base stations **20**, and the position (ground truth) of the terminal device **10** when the received signal strengths are measured.

[0150] Incidentally, the position of the terminal device **10** used as the ground truth in the training of the trained model that outputs the position of the terminal device **10** may be, for example, the position of the terminal device **10** estimated by executing the first estimation process using the received signal strengths at all the base stations **20** (i.e., the estimation result of the first estimation process), or may be the actual measured value measured using various sensors, the position input by the user, or the like.

[0151] As described above, the position estimation system **1** (position estimation device **30**) according to the embodiment includes the first estimation unit **33** which estimates the position of the terminal device **10** using the received signal strength at each of the multiple base stations **20** arranged in the predetermined space in which the terminal device **10** moves (i.e., the first communication status data concerning the communication status between the first wireless device moving in the space and each of the multiple second wireless devices arranged in the space), and the second estimation unit **34** which estimates the position of the terminal device **10** using the output of the trained model when the received signal strength is input to the trained model.

[0152] In this case, the received signal strength at each of the multiple base stations **20** is acquired at different intervals depending on the channel connecting the terminal device **10** with each of the multiple base stations **20**. In addition, the process executed by the first estimation unit **33** (first estimation process) or the process executed by the second estimation unit **34** (second estimation process) is selected based on the received signal strength at each of the multiple base stations **20**.

[0153] Furthermore, the trained model is built by using the training data including the received signal strength at each of the multiple base stations **20**, which is acquired (measured) when the terminal device **10** is located at a predetermined position (located at the same position) (i.e., the second communication status data related to the communication status between the terminal device **10** and each of the multiple base stations **20**). By learning from such training data, the trained model may be built to output the received signal strength at each of the multiple base stations **20** (for example, all the base stations **20**) by inputting the received signal strengths at at least some of the multiple base stations **20**. In addition, by learning from the training data including the received signal strength at each of the multiple base stations **20** and the estimation result of the first estimation process using the received signal strength, the trained model may be built to output the position (estimated value) of the terminal device **10** by inputting the received signal strengths at at least some of the multiple base stations **20**.

[0154] Incidentally, in the embodiment, the received signal strength at the first base station **20** (i.e., the first communication status data relating to the communication status between the terminal device **10** and some of the multiple base stations **20**) is measured (acquired) at a short interval (high frequency) using the communication channel. In addition, in the embodiment, the received signal strength at the second base station **20** (i.e., the first communication status data relating to the communication status between the terminal device **10** and the others of the multiple base stations **20**) is measured (acquired) at a long interval (low frequency) using the observation channel.

[0155] In the embodiment, the position of the terminal device **10** can be estimated at high speed with high accuracy by the above-described configuration.

[0156] More specifically, in the embodiment, for example, if the received signal strength is acquired such that the position of the terminal device **10** can be estimated with high accuracy, the first estimation process is selected and the position of the terminal device **10** can be estimated with high accuracy by executing the first estimation process.

[0157] More specifically, in the embodiment, for example, if the received signal strengths (i.e., the received signal strengths at a sufficient number of base stations **20**) are not acquired such that the position of the terminal device **10** can be estimated with high accuracy, the second estimation process is selected.

[0158] Since this second estimation process is executed without waiting until the received signals strength at a sufficient number of base stations **20** are acquired, faster process of estimating the

position of the terminal device **10** can be achieved.

[0159] Furthermore, in the second estimation process, by using the trained model built by using the above-described training data (for example, the trained model that can output the received signal strength at each of all the base stations **20** by inputting the received signal strengths at some base stations **20**), it is possible to achieve the accuracy in estimation of the position of the terminal device **10** at the same level as the first estimation process using the received signal strengths at a sufficient number of base stations **20**.

[0160] Incidentally, in the embodiment, it has been described that one terminal device **10** is provided for convenience. However, in an environment where multiple terminal devices **10** move in a predetermined space, the received signal strength at the base station **20** and the identifier for identifying the terminal device **10** may be acquired for each of the terminal devices **10**, and the position of each of the multiple terminal devices **10** may be estimated based on the received signal strength acquired for each of the terminal devices **10**.

[0161] In addition, the received signal strength acquired as described above is stored in the position estimation system **1** (for example, inside the position estimation device **30**) in association with the time at which the received signal strength is acquired. With this type of configuration, the first or second estimation process can be selected using the time at which the received signal strength is acquired.

[0162] More specifically, if the amount of change in the received signal strength at at least one base station **20** (for example, the first base station **20**) among the multiple base stations **20** is greater than or equal to a predetermined value, the second estimation process may be selected. In addition, if the interval of acquisition in the received signal strength at at least one base station **20** (for example, the first base station **20**) among the multiple base stations **20** is greater than or equal to a predetermined value, the second estimation process may be selected. Furthermore, if the base station **20** connected to the terminal device **10** over the communication channel is changed, the second estimation process may be selected. With such a configuration, it is possible to determine that the terminal device **10** has moved and to switch the first estimation process to the second estimation process.

[0163] In addition, if the number of base stations **20** from which the received signal strengths are acquired when the terminal device **10** is located at a predetermined position is less than a predetermined value, the second estimation process may be selected. With such a configuration, if the received signal strength at the required number of base stations **20** for the first estimation process is not acquired, the position of the terminal device **10** can be estimated with high accuracy using the second estimation process.

[0164] It has been described that the first or second estimation process is selected using the time at which the received signal strength is acquired. However, the time at which the received signal strength is acquired is also used when the first or second estimation process is executed. More specifically, the first or second estimation process is assumed to be executed using the received signal strength acquired during the period when the terminal device **10** is located at a predetermined position (i.e., a period from the time at which the terminal device **10** moves to the present time).

[0165] Furthermore, in the first estimation process, the position of the terminal device **10** may be estimated using the transmitted signal strength of the packets (signals) transmitted from the terminal device **10**. Incidentally, this transmitted signal strength may be acquired based on the type of terminal device **10** (for example, notebook PC, smartphone, or the like). With such a configuration, improvement in the accuracy of estimation of the position of the terminal device **10** can be expected.

[0166] In addition, in the first estimation process, the position of the terminal device **10** may be estimated based on the frequency band of the channel. With such a configuration, estimation of the position of the terminal device **10** in consideration of the fact that the propagation characteristics

differ depending on the frequency of the channel can be achieved.

[0167] Furthermore, in the first estimation process, if multiple antennas are built in each of the multiple base stations **20**, the received signal strength acquired for each of these antennas may be used. With such a configuration, when multiple base stations **20** (AP) including multiple antennas are arranged, the accuracy of the estimation of position of the terminal device **10** may be improved by using the information on the antenna used to observe the packet for measuring the received signal strength (in other words, by distinguishing the received signal strengths of the packets observed using different antennas).

[0168] In addition, in the first estimation process, the position of the terminal device **10** may be estimated using the transmitted signal strengths of the packets (signals) transmitted from the multiple base stations **20**. With such a configuration, the number of unknowns in the optimization problem expressed by the above-mentioned formula (2) can be reduced.

[0169] Furthermore, in the first estimation process, the position of the terminal device **10** is estimated using the position of each of the multiple base stations **20**. The position of each of the multiple base stations **20** may be acquired from the base stations **20** or, for example, may be held in advance in the position estimation device **30**. Furthermore, the position of each of the multiple base stations **20** may be estimated based on, for example, the received signal strength measured between the base stations **20**.

[0170] Incidentally, in the embodiment, the first estimation process estimates the position of the terminal device **10** by, for example, applying the received signal strength at each of the multiple base stations **20** to the minimization problem (a formula for radio wave propagation) expressed by formula (2). However, the process may use a formula other than that described in this embodiment.

[0171] Incidentally, the first estimation process has been described. However, the description of the first estimation process can also be applied to the same process as the first estimation process that is executed using the output of the trained model in the second estimation process.

[0172] In the embodiment, it has been described that the communication status data is the received power (data). However, the communication status data may be any data related to the communication status between the terminal device **10** and each of the multiple base stations **20** and, for example, a bit error rate or the like may be used as the communication status data.

[0173] In addition, in the embodiment, it has been described that the position estimation device **30** includes the units **31** to **35** shown in FIG. 4. However, at least some of these units **31** to **35** may be located outside the position estimation device **30**. Furthermore, the position estimation device **30** may be implemented by multiple devices, and the unit **31** to **35** may be arranged in different devices.

[0174] In the embodiment, estimating the position of the terminal device **10** such as a notebook PC used by a user moving in a predetermined space has been described. However, the embodiment can also be applied to a case of estimating the position of a wireless device arranged within the space, and can also be applied to a case where the terminal device **10** is an installation type device. For example, if the terminal device **10** is a desktop-type personal computer, it may be difficult to obtain a sufficient radio wave strength for position estimation over the communication channel immediately after startup. In such a situation, position estimation may be executed by scanning the observation channel, and then the position estimation may be executed by scanning the communication channel when a sufficient radio wave strength is obtained over the communication channel. In addition, for example, a configuration of estimating the position of a predetermined base station **20** or the like may be applied.

[0175] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their

equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

[0176] As regards the above-described embodiment, the following notes are further disclosed.

[0177] (1)

[0178] A position estimation system including a processor configured to: [0179] acquire first communication status data related to a communication status between a first wireless device and each of multiple second wireless devices; [0180] perform a first estimation process to estimate a position of the first wireless device using the acquired first communication status data; and [0181] perform a second estimation process to estimate the position of the first wireless device using an output of a trained model when the acquired first communication status data is input to the trained model, wherein [0182] the first communication status data is acquired at different intervals depending on a channel connecting the first wireless device with each of the multiple second wireless devices, [0183] the first estimation process or the second estimation process is selected based on the first communication status data, and [0184] the trained model is built by using the training data that includes second communication status data related to the communication status between the first wireless device and each of the multiple second wireless devices in a state in which the first wireless device is at a predetermined position.

[0185] (2)

[0186] The position estimation system of (1), wherein [0187] the first communication status data related to the communication status between the first wireless device and one of the multiple second wireless devices is measured at a short interval using a communication channel for executing wireless communication with the first wireless device, and [0188] the first communication status data related to the communication status between the first wireless device and the other of the multiple second wireless devices is measured at a long interval using an observation channel different from the communication channel.

[0189] (3)

[0190] The position estimation system of (1) or (2), wherein [0191] the first and second communication status data includes a received signal strength of the signal observed between the first wireless device and each of the multiple base stations.

[0192] (4)

[0193] The position estimation system of one of (1) to (3), wherein [0194] the first wireless device includes a terminal device used by a user, and the second wireless device includes a base station.

[0195] (5)

[0196] The position estimation system of one of (1) to (4), wherein the processor is configured to: [0197] when multiple first wireless devices move, acquire, in the first and second estimation processes, the first communication status data for each of the first wireless devices, and [0198] estimate a position of each of the multiple first wireless devices, based on the first communication status data acquired for each of the first wireless devices.

[0199] (6)

[0200] The position estimation system of (3), wherein [0201] the received signal strength included in the first communication status data is stored in the position estimation system in association with the time when the received signal strength is acquired.

[0202] (7)

[0203] The position estimation system of (6), wherein [0204] if an amount of change of the received signal strength of the signal observed between the first wireless device and at least one second wireless device among the multiple second wireless devices is larger than or equal to a predetermined value, the second estimation process is selected.

[0205] (8)

[0206] The position estimation system of (6), wherein [0207] if an interval of acquisition of the received signal strength of the signal observed between the first wireless device and at least one

second wireless device among the multiple second wireless devices is larger than or equal to a predetermined value, the second estimation process is selected.

[0208] (9)

[0209] The position estimation system of (6), wherein [0210] if the second wireless device connected to the first wireless device over the communication channel is changed, the second estimation process is selected.

[0211] (10)

[0212] The position estimation system of (6), wherein [0213] if the number of second wireless devices having the received signal strength acquired in a state in which the first wireless device is at a predetermined position is smaller than a predetermined value, the second estimation process is selected.

[0214] (11)

[0215] The position estimation system of (6), wherein [0216] the first wireless device is movable, and [0217] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using the received signal strength included in the first communication status data acquired after the first wireless device moves.

[0218] (12)

[0219] The position estimation system of (3), wherein [0220] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a transmitted signal strength of a signal transmitted from the first wireless device, and [0221] the transmitted signal strength is acquired based on information on a type of the first wireless device.

[0222] (13)

[0223] The position estimation system of one of (1) to (12), wherein [0224] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device based on a frequency band of the channel.

[0225] (14)

[0226] The position estimation system of one of (1) to (13), wherein [0227] if multiple antennas are built in each of the multiple second wireless devices, the processor is configured to acquire the first communication status data for each of the antennas.

[0228] (15)

[0229] The position estimation system of one of (1) to (14), wherein [0230] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a transmitted signal strength of a signal transmitted from each of the multiple second wireless devices.

[0231] (16)

[0232] The position estimation system of (3), wherein [0233] each of the multiple second wireless devices is configured to instruct the first wireless device to scan the signal and acquire a received signal strength of a signal from the first wireless device as the scanning result, and [0234] the processor is configured to acquire the received signal strength from each of the multiple second wireless devices.

[0235] (17)

[0236] The position estimation system of one of (1) to (16), wherein [0237] the trained model is built to output fourth communication status data on a communication status between the first wireless device and each of the multiple second wireless devices by inputting third communication status data on a communication status between the first wireless device and at least some of the multiple second wireless devices, by learning from training data including the second communication status data.

[0238] (18)

[0239] The position estimation system of one of (1) to (16), wherein [0240] the trained model is built to output the position of the first wireless device by inputting third communication status data

on a communication status between the first wireless device and at least some of the multiple second wireless devices, by learning from training data including the second communication status data and a result of the first estimation process using the second communication status data.

[0241] (19)

[0242] The position estimation system of one of (1) to (18), wherein [0243] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a position of the second wireless device acquired from each of the multiple second wireless devices.

[0244] (20)

[0245] The position estimation system of one of (1) to (18), wherein [0246] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using the position of each of the multiple second wireless devices held in advance.

[0247] (21)

[0248] The position estimation system of one of (1) to (18), wherein [0249] the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a position of each of the multiple second wireless devices, which is estimated based on fifth communication status data indicating a communication status among the multiple second wireless devices.

[0250] (22)

[0251] The position estimation system of (3), wherein [0252] the processor is configured to estimate, in the first estimation process, the position of the first wireless device by applying the received signal strength included in the acquired first communication status data to a calculation formula on radio wave propagation.

Claims

1. A position estimation system comprising a processor configured to: acquire first communication status data related to a communication status between a first wireless device and each of multiple second wireless devices; perform a first estimation process to estimate a position of the first wireless device using the acquired first communication status data; and perform a second estimation process to estimate the position of the first wireless device using an output of a trained model when the acquired first communication status data is input to the trained model, wherein the first communication status data is acquired at different intervals depending on a channel connecting the first wireless device with each of the multiple second wireless devices, the first estimation process or the second estimation process is selected based on the first communication status data, and the trained model is built by using the training data that includes second communication status data related to the communication status between the first wireless device and each of the multiple second wireless devices in a state in which the first wireless device is at a predetermined position.
2. The position estimation system of claim 1, wherein the first communication status data related to the communication status between the first wireless device and one of the multiple second wireless devices is measured at a short interval using a communication channel for executing wireless communication with the first wireless device, and the first communication status data related to the communication status between the first wireless device and the other of the multiple second wireless devices is measured at a long interval using an observation channel different from the communication channel.
3. The position estimation system of claim 2, wherein the first and second communication status data includes a received signal strength of the signal observed between the first wireless device and each of the multiple base stations.
4. The position estimation system of claim 3, wherein the first wireless device includes a terminal device used by a user, and the second wireless device includes a base station.

5. The position estimation system of claim 1, wherein the processor is configured to: when multiple first wireless devices move, acquire, in the first and second estimation processes, the first communication status data for each of the first wireless devices, and estimate a position of each of the multiple first wireless devices, based on the first communication status data acquired for each of the first wireless devices.
6. The position estimation system of claim 3, wherein the received signal strength included in the first communication status data is stored in the position estimation system in association with the time when the received signal strength is acquired.
7. The position estimation system of claim 6, wherein if an amount of change of the received signal strength of the signal observed between the first wireless device and at least one second wireless device among the multiple second wireless devices is larger than or equal to a predetermined value, the second estimation process is selected.
8. The position estimation system of claim 6, wherein if an interval of acquisition of the received signal strength of the signal observed between the first wireless device and at least one second wireless device among the multiple second wireless devices is larger than or equal to a predetermined value, the second estimation process is selected.
9. The position estimation system of claim 6, wherein if the second wireless device connected to the first wireless device over the communication channel is changed, the second estimation process is selected.
10. The position estimation system of claim 6, wherein if the number of second wireless devices having the received signal strength acquired in a state in which the first wireless device is at a predetermined position is smaller than a predetermined value, the second estimation process is selected.
11. The position estimation system of claim 6, wherein the first wireless device is movable, and the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using the received signal strength included in the first communication status data acquired after the first wireless device moves.
12. The position estimation system of claim 3, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a transmitted signal strength of a signal transmitted from the first wireless device, and the transmitted signal strength is acquired based on information on a type of the first wireless device.
13. The position estimation system of claim 1, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device based on a frequency band of the channel.
14. The position estimation system of claim 1, wherein if multiple antennas are built in each of the multiple second wireless devices, the processor is configured to acquire the first communication status data for each of the antennas.
15. The position estimation system of claim 1, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a transmitted signal strength of a signal transmitted from each of the multiple second wireless devices.
16. The position estimation system of claim 3, wherein each of the multiple second wireless devices is configured to instruct the first wireless device to scan the signal and acquire a received signal strength of a signal from the first wireless device as the scanning result, and the processor is configured to acquire the received signal strength from each of the multiple second wireless devices.
17. The position estimation system of claim 1, wherein the trained model is built to output fourth communication status data on a communication status between the first wireless device and each of the multiple second wireless devices by inputting third communication status data on a communication status between the first wireless device and at least some of the multiple second

wireless devices, by learning from training data including the second communication status data.

18. The position estimation system of claim 1, wherein the trained model is built to output the position of the first wireless device by inputting third communication status data on a communication status between the first wireless device and at least some of the multiple second wireless devices, by learning from training data including the second communication status data and a result of the first estimation process using the second communication status data.

19. The position estimation system of claim 1, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a position of the second wireless device acquired from each of the multiple second wireless devices.

20. The position estimation system of claim 1, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using the position of each of the multiple second wireless devices held in advance.

21. The position estimation system of claim 1, wherein the processor is configured to estimate, in the first and second estimation processes, the position of the first wireless device, using a position of each of the multiple second wireless devices, which is estimated based on fifth communication status data indicating a communication status among the multiple second wireless devices.

22. The position estimation system of claim 3, wherein the processor is configured to estimate, in the first estimation process, the position of the first wireless device by applying the received signal strength included in the acquired first communication status data to a calculation formula on radio wave propagation.
